🔑 **Answer Key — Teacher Copy · Lesson 6-10**

MSTAR PRACTICE WORKSHEET

Lesson 6-10: Divide Mixed Numbers

Grade 6 Mathematics · Standard 6.NOS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

What is 1 1/2 ÷ 3/4?

- A 1 1/8
- **B 2**
- C 3/4
- D 9/8

1 1/2 = 3/2. Then 3/2 ÷ 3/4 = 3/2 × 4/3 = 12/6 = 2.

Question 2 SELECTED RESPONSE — CORRECT: C

Convert 3 2/5 to an improper fraction.

- A 6/5
- B 15/5
- **C 17/5**
- D 32/5

3 × 5 = 15, then 15 + 2 = 17. So 3 2/5 = 17/5.

Question 3 SELECTED RESPONSE — CORRECT: C

What is 2 1/4 ÷ 3/4?

- A 1 1/2
- B 2 1/3
- C 3
- D 9/16

> 2 1/4 = 9/4. Then 9/4 ÷ 3/4 = 9/4 × 4/3 = 36/12 = 3.

Question 4 SELECTED RESPONSE — CORRECT: D

What is 3 1/2 ÷ 1/4?

- A 3 1/8
- B 7/8
- C 13/4
- D 14

> 3 1/2 = 7/2. Then 7/2 ÷ 1/4 = 7/2 × 4/1 = 28/2 = 14.

Question 5 SELECTED RESPONSE — CORRECT: C

A baker has 2 1/4 cups of butter. Each batch of cookies needs 3/4 cup of butter. How many batches can the baker make?

- A 1 1/2 batches
- B 2 batches
- C 3 batches
- D 6 batches

> Convert 2 1/4 = 9/4. Then 2 1/4 ÷ 3/4 = 9/4 × 4/3 = 36/12 = 3 batches.

Question 6 SELECTED RESPONSE — CORRECT: B

Mission 1: The stakeout shift is 4 1/2 hours long. Each agent covers a 1 1/2-hour watch. How many agent watches fill one shift?

- (A) 2 1/2
- (B) 3
- (C) 4 1/2
- (D) 6

> 4 1/2 = 9/2 and 1 1/2 = 3/2. So 9/2 ÷ 3/2 = 9/2 × 2/3 = 18/6 = 3 watches.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

What is the value of 3 3/4 ÷ 1 1/2?

- (A) 1 1/2
- (B) 2 1/2
- (C) 2
- (D) 5 5/8

> Convert both mixed numbers to improper fractions: 3 3/4 = 15/4 and 1 1/2 = 3/2. Then keep, change, flip: 15/4 × 2/3 = 30/12 = 5/2 = 2 1/2.

- **A:** That is the divisor itself, not the quotient. Keep 15/4, change to multiply, flip 3/2 to 2/3.
- **C:** The whole numbers were divided and the fractions dropped. Convert to 15/4 and 3/2 first.
- **D:** That multiplies without flipping. Change the divisor 3/2 to 2/3 before multiplying.

Part B — correct answer: C

Which reasoning shows why your answer to Part A is correct?

- (A) Convert 3 3/4 by multiplying 3 × 4 and keeping the denominator, giving 12/4, then 12/4 × 2/3 = 2.
- (B) Ignore the whole numbers and divide only the fraction parts: 3/4 ÷ 1/2 = 3/4 × 2/1 = 6/4 = 1 1/2.
- (C) Convert to improper fractions, 15/4 and 3/2, then keep, change, flip: 15/4 × 2/3 = 30/12 = 5/2 = 2 1/2.
- (D) Convert to improper fractions and multiply straight across without flipping: 15/4 × 3/2 = 45/8 = 5 5/8.

> Converting 3 3/4 requires 3 × 4 + 3 = 15 in the numerator, giving 15/4. After keep, change, flip the product is 30/12, which simplifies to 2 1/2.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Skipping the Conversion

Tyrone divided 2 1/2 ÷ 1/4. His work was: keep 2 1/2, change ÷ to ×, flip 1/4 to 4/1. Then he multiplied only the whole number, 2 × 4 = 8, and carried the 1/2 along, so he answered 8 1/2.

Explain Tyrone's misconception and give the correct quotient.

Model answer: Tyrone never converted the mixed number, so the 1/2 was never multiplied by 4. Convert first: 2 1/2 = 5/2. Then keep, change, flip: 5/2 × 4/1 = 20/2 = 10. The quotient is 10.

2 points	Student explains that 2 1/2 must be converted to the improper fraction 5/2 before multiplying so that the half is included, and gives the quotient 10.
1 point	Student gives the quotient 10 but does not explain the missing conversion to an improper fraction.
0 points	Student agrees with Tyrone that the quotient is 8 1/2 or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A detective has a rope that is 5 1/4 feet long. She needs to cut it into pieces that are each 3/4 foot for tying evidence tags. How many pieces can she cut? Write the equation, show each step (convert, keep-change-flip, multiply, simplify), and verify your answer.

Model answer: The equation is 5 1/4 ÷ 3/4, so convert 5 1/4 = (5 × 4 + 1)/4 = 21/4, then Keep-Change-Flip: 21/4 × 4/3 = 84/12 = 7 pieces. Multiplying back checks it: 7 × 3/4 = 21/4 = 5 1/4 feet of rope.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.